Capstone Project

The Elements of Statistical Learning: February – June 2026

The capstone project serves as a pivotal assignment, bringing together the knowledge and skills you've developed throughout your studies. It involves an in-depth exploration of data of your choice to identify a real-world challenge and prompts you to ask: How can I design a machine learning system that effectively resolves this issue — for myself or for a company dealing with it?

1. Project Outline

You start by choosing a well-defined problem that can be addressed using methods covered in the course or through independent study. Acquire and clean an appropriate dataset, then select two machine-learning scenarios from the list below (or propose an equivalently challenging, tutor-approved alternative). Present your results in a polished Shiny application accompanied by a one-page executive summary. Provide full code and code documentation with reusability in mind.

During the examination, demonstrate your ability to explain and defend each step of your data processing and model selection. Conclude with a clear rationale for why your chosen solution effectively addresses the problem.

2. Selecting Data

The goal is to work with genuinely raw data and avoid relying on existing code or pre-solved workflows[1]. You will present your chosen dataset during the practical session, where it must be approved by your tutor. Projects are completed in teams of two, and each dataset may be used only twice across all teams. To prevent overlap, register your selected dataset in the shared spreadsheet and verify that it does not conflict with another group's choice[2].

3. Hypothesis-Driven Project

After collecting your data, the next step is to define a focused hypothesis that matches your research questions. Because your **models must involve several dozen parameters** (ideally >30), simple visual EDA will not be sufficient at the outset, and some tasks will require revisiting lecture material or consulting additional sources. Your hypothesis should be formulated to guide modelling decisions in these high-dimensional settings and ensure that each analytical step follows a clear, defensible rationale.

You start by selecting two scenarios from the list below and consulting them with your tutor. Some options can be linked, so choosing one may require thinking about the related one as well.

1. Compare a regression or classification setup across three statistical or machine-learning methods and two feature-space partitioning approaches. Identify which features remain informative across

[1] Public data repositories with integrated solutions, such as Kaggle, are not allowed. Instead, turn your attention to local or global open data repositories, such as data.bratislava.sk, www.clinpgx.org, etc.

[2] https://docs.google.com/spreadsheets/d/1UQjKAs4CCNZiv9dQqqf6t3XTHkLcK7t-QLm3HFzQkPA/edit?usp=sharing

both method families, discuss the implications for explainability, and outline the benefits each approach offers.

2. Compare parametric and non-parametric models for your chosen problem, selecting at least three methods from each group. Evaluate which class offers stronger explainability and reproducibility and justify why.
3. Extend your work by comparing algorithmic and embedded feature-selection methods for the same regression or classification task. This task may require some research on your side, as not all necessary information will be taught through the course. Report how many features you ultimately retain, and which features lose statistical significance as you add or remove predictors.
4. Following a standard classification workflow — whether regression-based or tree-based — you may also visualize labelled data using heatmaps or tree diagrams. Compare how these visualizations differ from the underlying models and to what extent they capture model-based classification. Specify which parameters you tuned to make the heatmaps or trees to align as closely as possible with the model behavior and discuss the key similarities and differences between the two approaches.
5. If your dataset is highly multidimensional, you may apply a dimensionality-reduction technique before running regression or classification with the methods discussed in class. Compare approaches such as PCA, LDA, or related techniques, and explain how each transforms the feature space. Evaluate whether explainable machine learning remains feasible after this reduction and what information is lost in the process.

4. First Steps

Dive into the data, analyse basic stats, and check if its features, format, quality, and quantity are suitable for testing your hypothesis using machine/statistical learning. If things don't line up, take the time to get new data or preprocess the data to meet the requirements of your chosen method.

Maintain full experimental reproducibility by documenting every step and explaining the reasoning behind each decision. Use Python or the Tidyverse dialect of R for basic programming. Make sure that your submission is fully runnable and reproducible: use R Markdown or Quarto for R, and Jupyter notebooks for Python, and include all data required to execute the project without modification.

5. Model Fitting, Requirement Analysis and Performance Evaluation

Based on your chosen scenario and method chosen, perform requirement analysis, data budgeting, function fitting and assess model performance. Explain your rational for selecting a selected specific model and build your strategy to compare them. Carefully document each step of the process, emphasizing the formulation of the model and avoiding overcomplication.

6. Visualization

Use visualizations selectively, mainly to justify your choices and support your conclusions. Avoid bloated, unfocused EDA. Be deliberate and critical about every plot you include.

7. Deliverables

Prepare your presentation using a combination of one-pager summary in lazy language, technical project documentation in R markdown or Jupyter Notebook (without length restriction, just make sure that you have every key point explained) and a Shiny application for your oral presentation.

1. Create executive summary: Provide a short overview of your domain background, main findings, and proposed solution. Clearly justify why you chose your models and highlight the key insights that emerged when comparing them with the alternatives required in the assignment. Present your

conclusions in clear, non-technical language so they are understandable to a broad audience. Submit the summary as a Word document or PDF. **Write this summary entirely yourself without using LLM-generated text.**

2. Create complete project documentation in R Markdown or Jupyter Notebooks. Know your code, explain each step clearly so a technical reader can follow your workflow. Comment on model choices, parameter settings, inputs, and outputs, and provide a detailed interpretation of the results.
3. For the oral presentation**,** create a Shiny application that showcases the data and fitted models to the client. The application should feature a user interface that enables data loading, function fitting, and dynamic model parameter adjustments. Any modifications made should automatically update the model. A working example of such an application was demonstrated in the last lecture, but you are free to explore Shiny's feature demos[3] for additional inspiration.

8. Generative AI and Cheating

Generative AI may be used only in limited ways. It can support the structure of your summary, but generating the actual text of your reports is not allowed. You may use it to assist with coding, but you must fully understand every function you include in your workflow. Lack of understanding will result in zero points. Reusing previously published code or code from your fellow students is considered cheating.

9. Teamwork

You will collaborate in teams of two. Open and effective communication with your partner is essential when dividing the workload, assigning tasks, and ensuring responsibilities are clearly understood. Take the time to plan the project's timeline, establish checkpoints, and monitor progress to keep the work on schedule. Actively support each other throughout the process, address challenges together, and stay aligned with the shared goal of delivering the project successfully and on time.

10. Scoring Rubic

		Points
One-pager	Write an executive summary that highlights the project's aims, main results, and overall conclusions in clear, high-level language suitable for a lay audience.	5
Project documentation	Document your project from start to finish, covering data selection and preparation, the modelling scenario you chose and why it fits the data, your analysis, comparisons, and visualizations. Produce all required deliverables at a quality suitable for reproducible and maintainable work. Include visualizations where they meaningfully support your conclusions.	30
Project presentation	Develop a Shiny application that integrates your implemented models, helps users explore the data, and lets them interactively see how model outputs change when parameters are adjusted.	10
Free questions	You will be asked three questions to demonstrate a clear understanding of your code, your data, and the theoretical concepts needed for modelling.	15
	Total:	**60**

There are 60 total points available, and you must earn **at least 40** (66%) to pass the assignment.

[3] https://shiny.posit.co/r/gallery/#feature-demos